and should i use also tsne ?

Short answer: No, you should NOT use t-SNE in your main pipeline for clustering + surrogate modeling.

Here's why, and how to handle it properly:

Key Problem with t-SNE

Issue	Why it matters
Non-deterministic [1]	Same data → different output each run; reproducibility issues
Cannot transform test data [1]	t-SNE doesn't have a `fit_transform()` that works on new data; you can't apply it to test sets
Not for ML pipelines [1]	**t-SNE is primarily a visualization tool**, not for downstream modeling
No global structure [1] [2]	Distorts inter-cluster distances; only preserves local neighborhoods
Computationally expensive [3]	Slower than UMAP, especially for large datasets

When t-SNE IS Appropriate

✅ **Use t-SNE only for:**

- **Visualization purposes only** (e.g., 2D plots in your report)
- **Exploratory analysis** to understand data structure
- **Comparing with UMAP for visualization quality** (in a supplementary section)

❌ **Do NOT use t-SNE for:**

- Dimensionality reduction before clustering
- Input to surrogate models
- Any pipeline where you need to transform new/test data

Revised Recommendation for Your Report

Main Pipeline (for clustering + surrogate modeling):

```
Dimensionality Reduction: PCA or UMAP only
↓
Clustering: KMeans, HDBSCAN, Agglomerative, Spectral, GMM
↓
Cluster Labels
↓
Visualization: PCA 2D, UMAP 2D (for display)
↓
Surrogate Model: RF, DT, XGBoost
```

Optional Supplementary Section (visualization comparison only):

```
Add a small section comparing PCA 2D vs UMAP 2D vs t-SNE 2D
for visualization quality (not for modeling)
```

Why UMAP is Better Than t-SNE for Your Case

Criterion	UMAP	t-SNE
Preserves global structure	✅ Yes (partially)	❌ No
Faster runtime	✅ Yes	❌ Slower
Deterministic	✅ Yes (with seed)	❌ No
Supports test transformation	✅ Yes	❌ No
Better for clustering	✅ Yes	⚠ Limited
Visualization quality	✅ Good	✅ Good (similar)

Research shows: **UMAP produces similar results to t-SNE but faster** and with better global structure preservation. [4] [5]

Final Suggestion

For your **formal report** comparing dimensionality reduction + clustering + surrogate modeling:

Option A (Most Conservative):

- **Use only PCA and UMAP** for your main 30 pipelines
- Add a **small supplementary figure** showing t-SNE 2D visualization for comparison (not metrics)

Option B (More Comprehensive):

- **Use only PCA and UMAP** for the full 30-pipeline comparison
- In your **discussion/limitations section**, mention: *"t-SNE was evaluated for visualization only but excluded from pipelines due to its inability to transform test data and non-deterministic behavior"*

This keeps your **methodology rigorous and reproducible** while acknowledging t-SNE's role as a visualization tool. [1] [2] [5]

Bottom line: Stick with **PCA + UMAP** for your main comparison. Use t-SNE only if you want an extra visualization figure in your appendix (not for modeling).

⁂

1. https://www.reddit.com/r/datascience/comments/wy1rmk/pca_vs_umap_vs_tsne_on_a_very_layman_level_what/
2. https://biostatsquid.com/pca-umap-tsne-comparison/
3. https://aurigait.com/blog/blog-easy-explanation-of-dimensionality-reduction-and-techniques/
4. https://www.youtube.com/watch?v=aBUuNHt3YsA
5. https://mcpanalytics.ai/articles/t-sne-vs-pca-vs-umap-which-reveals-true-clusters
6. https://medium.com/@hagarwaa/comparative-analysis-of-pca-t-sne-and-umap-a-dimensionality-reduction-perspective-3f9b82064df6
7. https://www.kaggle.com/general/560484
8. https://www.bioinformatics.babraham.ac.uk/training/10XRNASeq/Dimension Reduction.pdf
9. https://www.youtube.com/watch?v=xKWPe9jxPG8
10. https://carnotresearch.medium.com/understanding-dimensionality-reduction-pca-vs-t-sne-vs-umap-vs-fit-sne-vs-largevis-vs-laplacian-13d0be9ef7f4